

PRODUCT REQUIREMENTS DOCUMENT

Interactive Ideas

Game Mechanics Design Specification

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"The quest is not a metaphor for the work. The work is the quest."

1. Overview & Project Types

Interactive Ideas is a gamified productivity platform for Indian college students. The core metaphor is a 2D arcade-sidescroller RPG where real intellectual and creative work is the gameplay. Users build real projects — ventures, academic papers, lab experiments, creative pieces — inside a quest-world that treats each stage of the work as a stage of a journey through a high-fantasy landscape.

When a user clicks Create Project, they choose from four archetypes. Each has its own world skin, stage structure, monster roster, and AI quality metric.

Project Type	Template	Stages	Quality Metric	Direction
Venture	A startup or business idea. Stages from Ideation through Scale.	8	Valuation Score (INR proxy)	Always increases
Academic	A research paper or essay. Topic through Submission.	6	JIF Score (Impact Factor proxy)	Always increases
Lab (Experimental)	Hands-on experiment, prototype, or technical build.	7	p-value (statistical significance)	Decreases toward 0.05
Creative	Writing, music, visual art, film, or any creative form.	6	Fan Score (audience reach proxy)	Always increases

DESIGN PRINCIPLE

The four templates share the same underlying checkpoint/boss/corruption engine. What differs is the world skin (visual setting per stage), the stage names and lore, the monster roster, and the AI quality signal used to scale boss difficulty. All game systems are template-agnostic unless explicitly noted.

2. Brief, AI Tagging & User Matching

2.1 Project Brief Flow

After selecting a project type, the user writes a free-text brief describing the idea. The brief populates the project's description field directly — the brief IS the content. Recommended length: 150-600 words.

Flow: Select Type → Write Brief → AI Suggests Title + Tags → User Confirms/Edits → Matching + World Launch

2.2 AI Title & Tag Generation

On brief submission, the backend calls the AI model (open-weight for free tier, frontier for Explorer Pro). The AI returns:

- Title — 3-7 word project name, specific and memorable
- Skill Tags — 2-5 tags from a fixed taxonomy (e.g. Machine Learning, Product Design, Financial Modelling)
- Industry Tags — 2-4 tags from a fixed taxonomy (e.g. EdTech, Climate, Consumer)

The user sees AI suggestions and can freely override any or all before confirming. Tags are selected from autocomplete dropdowns backed by a fixed taxonomy. Custom tags are not permitted — all tags must exist in the platform taxonomy to enable matching.

2.3 Collaborator Matching Algorithm

Once a project is created, potential collaborators are surfaced in the project Recruit panel, ranked by the following tiered scoring algorithm:

Priority Tier	Match Condition	Base Score	Modifier
Tier 1 (Highest)	Skill tags AND industry tags overlap	+100 base	+15 per skill match, +10 per industry match
Tier 2	Skill tags overlap only	+60 base	+15 per additional skill match
Tier 3	Industry tags overlap only	+30 base	+10 per additional industry match
Tier 4 (Lowest)	No tag overlap — random serendipity	+0-10 random	—
Modifier: Experienced	User level >= 10	+5	Applied after tier score
Modifier: Active Now	User currently online	+8	Applied after tier score

Priority Tier	Match Condition	Base Score	Modifier
Modifier: Prior Collab	Already collaborated before	-20	Prevents over-suggesting same people

UX NOTE

The matching panel is available throughout the project lifecycle — not just at creation. Tier labels are shown to the project owner (e.g. Skill + Industry Match) but not exposed to the potential collaborator to avoid creating a merit hierarchy in the invitation experience.

3. Persona & Character System

Once the project is confirmed and the world launches, the user selects an animated pixel persona — the avatar that inhabits and navigates the quest world. Each persona is a pixel-art sprite with an idle animation loop and walk/run cycle. Personas are project-specific; a user may choose a different persona per project.

3.1 Predefined Persona Pool — 10 Sprites

The platform ships with 10 predefined pixel personas. Each is 32x48px at native resolution, scaled 3x for display (96x144px rendered). The pool represents mixed gender and diverse visual heritage.

Persona Name	Archetype	Class
The Arcanist	Wizard	Male
The Ranger	Scout	Female
The Alchemist	Scholar	Nonbinary
The Artisan	Creator	Female
The Drifter	Rogue	Male
The Oracle	Seer	Female
The Engineer	Builder	Male
The Healer	Cleric	Female
The Pathfinder	Explorer	Nonbinary
The Sage	Mentor	Male

3.2 Custom Persona Options

Option A — Character Creator (Explorer Pro)

An in-app pixel sprite builder. Users pick a base body, skin tone from a curated set, hair style, and outfit. Results in a unique sprite rendered with the same animation rig as the predefined pool.

Option B — Photo-to-Pixel Conversion (Explorer Pro)

User uploads a photo; a server-side pipeline converts it to a 32x48px pixel portrait using palette-reduction and Bayer dithering, then applies the standard animation rig automatically.

IMPLEMENTATION NOTE — PHOTO TO PIXEL PIPELINE

Recommended approach: resize photo to 32x48, quantise to a 16-colour palette derived from the platform design system, apply ordered Bayer dithering, then composite onto a transparent sprite sheet. The animation rig (idle bob, walk frames) is applied procedurally using bone-mapped offset rules tied to the head/torso/leg

regions. Implementable client-side using Canvas API with a WebGL post-processing pass, or server-side with Pillow + a custom dithering script.

4. World Map & Stage Architecture

Each project renders as a 2D fixed-path sidescroller world — a snake-path overworld map (Phaser 3 canvas, mounted inside React) where checkpoints are nodes along a road through multiple distinct stage environments. The visual metaphor is a classic JRPG overworld condensed into a single scrollable canvas.

4.1 Stage Table — Venture Template (8 Stages)

#	Stage Name	Setting / Biome	Stage Monster	CPs
S1	Ideation	The Village	Fog of Vagueness	4
S2	Research	The Forest	Pathwarden Wraith	5
S3	Validation	The Arena	Advocate of Comfortable Lies	4
S4	Offer Design	The Artisan's Quarter	Unfinished Golem	5
S5	Build & Deliver	The Mine	Collapse Specter	6
S6	Launch	The Harbour	Harbourmaster of Hesitation	3
S7	Iteration	The Crossroads Town	Babel Merchant	4
S8	Scale	The Capital	Iron Bureaucrat	5

4.2 Other Template Stage Structures

Academic — 6 Stages

Stage	Setting	Stage Monster
Topic & Question	Ancient Library	Librarian of Lost Questions
Literature Review	The Ruins	Keeper of Incomplete Records
Methodology	Cartographer's Tower	Cartographer of Crooked Maps
Writing & Drafting	The Scriptorium	Blank Page Wraith
Review & Revision	Council Chamber	Councillor of False Consensus
Submission & Publication	Grand Archive	Gatekeeper of Unearned Entry

Lab (Experimental) — 7 Stages

Stage	Setting	Stage Monster
Brief & Question	The Observatory	Mirage Lens
Background Research	Ancient Library	Librarian of Lost Questions

Stage	Setting	Stage Monster
Design & Planning	Cartographer's Tower	Cartographer of Crooked Maps
Build & Execute	The Forge	Saboteur of the Forge
Test & Evaluate	Alchemist's Laboratory	Alchemist of Wishful Results
Iterate & Refine	The Crossroads Town	Babel Merchant
Document & Present	The Grand Hall	Silencer of Findings

Creative — 6 Stages

Stage	Setting	Stage Monster
Concept & Inspiration	The Sacred Grove	Silence That Smothers
References & Influences	Gallery of Echoes	Curator of Derivative Ghosts
Drafting & Creation	The Wilderness	Beast of the Unfinished
Feedback & Critique	Village Square	Crowd of False Validation
Refinement & Polish	Artisan's Workshop	Perfectionist's Spectre
Release & Sharing	The Harbour	Harbourmaster of Hesitation

4.3 Map Rendering Rules

- The overworld map scrolls horizontally left-to-right. Each stage occupies a defined horizontal band with a distinctive biome skin.
- Checkpoints within a stage are rendered as nodes on a snake path within that biome band.
- The persona sprite floats above the current active checkpoint node with an idle animation.
- Completed checkpoints display their completion state: standard glow vs. gold burst. Locked checkpoints are visually dimmed with a lock icon.
- The stage boss appears as a large silhouette at the far end of each stage band, with a health/corruption bar beneath it.
- The Super Boss (randomly assigned per project) casts a faint shadow across the entire map from off-screen, becoming fully rendered only when all stage bosses are defeated.

5. Checkpoint & Task Mechanics

5.1 The Three-Task Gate

Every checkpoint contains exactly 3 tasks: one Easy (T1), one Medium (T2), and one Stretch (T3). The user must complete any 2 of 3 to advance past the checkpoint. Completing all 3 earns a Gold Checkpoint — the highest quality state.

TASK POINT WEIGHTS
T1 Easy = 20% of checkpoint points | T2 Medium = 20% | T3 Stretch = 35% | Gold Bonus (all 3) = additional 25%. The asymmetric weighting incentivises the stretch task — doing T1+T3 yields more points than T1+T2, even though both advance the checkpoint.

5.2 Task Tool Types

Each task is associated with a specific tool type that determines how the user submits their work:

Tool Type	Description
Write	Rich text editor with markdown support, AI assist, and word count
Table	Structured spreadsheet-style table with labelled columns
Map / Canvas	Freeform whiteboard — post-its, diagrams, arrows, images
Survey	Create, distribute, and collect surveys. Responses visualised.
Poll	Quick 2-4 option polls broadcastable to community
Link	URL input with auto-preview and annotation fields
Upload	File upload: PDF, PPT, XLS, DOC, images, video, audio
Self-report	Guided text form for personal confirmations and reflections
Journal	Private running log; entries selectively shareable
Kanban	Task management board with custom columns
Calendar	Timeline and milestone scheduling tool

5.3 Checkpoint Completion States

State	Condition	Visual Treatment	Boss Impact
Partial	1/3 tasks done	Node glows amber. Partial crack in seal. Persona paces.	No boss hit. Corruption reduced slightly in proximity.
Standard Clear	2/3 tasks done	Seal shatters, gate opens, crossing animation plays. Node lit with	One hit on stage monster. Corruption meter -12%.

State	Condition	Visual Treatment	Boss Impact
		white-blue glow.	
Gold Clear	3/3 tasks done	Seal explodes in gold particles. Gate transforms to gilded arch. Gold crown above node. Community feed notification.	Two hits on monster. Corruption meter -25%. Gold marker permanent.

5.4 Stage Final Checkpoint States

- Partial Stage — At least half the stage CPs completed but final CP not gold. Monster retreats rather than dying. One residual path element remains.
- Stage Clear — Final CP standard-completed. Monster slain. Stage biome fully restores visually. Stage Clear card animation plays (arcade-style banner drop).
- Gold Stage — Final CP gold-completed. Unique monster slay animation triggers. Stage biome transforms — colour-floods, particles, landscape elevation change.

5.5 Project Final Checkpoint States

- Partial Project — Some stages not gold. Super Boss retreats. World map shows mixed completion states. Monument placed but cracked.
- Project Complete — All stages cleared (2/3 or better). Super Boss slain. Full world map celebration. Monument placed: solid stone, persona engraved.
- Project Perfect — All final stage CPs gold. Super Boss slay triggers unique defeat animation. Monument transforms to gold. Legendary completion badge awarded.

6. Boss & Corruption System

6.1 The Corruption Mechanic

Each project has a single, persistent Corruption Meter — a bar displayed in the HUD at all times. It represents the accumulated entropy working against the project.

CORRUPTION RULES
+5% per day of inactivity (capped at 80%) | +10% for a checkpoint left at 1/3 tasks for more than 5 days | -12% for standard checkpoint clear (2/3) | -25% for gold checkpoint clear (3/3) | -5% for submitting a Contribution update | Corruption visual animations vary at 25%, 50%, 75%, 90% thresholds | At 100%: Super Boss enters the map with a cinematic animation. No data is lost but the visual state is alarming.

6.2 Corruption Visual Thresholds

Threshold	World Visual	HUD / Audio
0-25%	Baseline world. Faint purple tinting on horizon. Boss silhouette barely visible.	Normal ambience.
25-50%	Corruption overlay creeps from map edges. Ahead checkpoints slightly dimmed. Boss solidifies.	Low rumble begins. Ambient pitch shifts darker.
50-75%	World desaturates noticeably. Boss begins slow movement. Purple vignette on HUD border.	Corruption hum louder. Occasional distant thunder.
75-90%	Map heavily corrupted. Boss fully visible. Active checkpoint shakes. HUD border pulses.	Urgent drone. Boss-specific ambient SFX begins.
90-100%	Full corruption. Boss enters foreground. Screen edge cracks. Flash warning every 30s.	Dramatic boss arrival SFX. Persistent alarm undertone.

6.3 Super Boss Pool — 12 Bosses (One Randomly Assigned Per Project)

#	Boss Name	Type	Represents	Defeated By
01	The Unraveller	Ancient Void Serpent	Doubt and loss of direction	Weaving the final stage into a coherent whole
02	The Pale Architect	Undead Perfectionist Titan	Paralysis and perfectionism	Shipping something imperfect; completion despite flaws
03	The Hollow King	Spectral Sovereign	Loss of purpose — doing the work without knowing why	Reconnecting to the original impulse

#	Boss Name	Type	Represents	Defeated By
04	The Thornwarden	Ancient Forest Colossus	Bureaucracy and friction	Each gold checkpoint clears a path it can't regrow
05	The Mirror Witch	Illusionist Sorceress	Confirmation bias and self-deception	Each validation or feedback task breaks a mirror
06	The Ashen Drake	Fire Dragon of Entropy	Abandonment and inertia	Consistent forward motion — it can't burn what's being built
07	The Tide Caller	Oceanic Leviathan	Distraction and scope creep	Each prioritisation or synthesis task drains the flood
08	The Gravemind	Necromantic Hive	Fear of failure and past attempts	Each completed checkpoint buries a corpse permanently
09	The Rusted Oracle	Corrupted Mechanical Prophet	Imposter syndrome	Each original insight silences one of its voices
10	The Wraith Council	Parliament of Failed Founders	Decision paralysis and committee thinking	Each decisive checkpoint dismisses one councillor
11	The Stonecaller	Mountain Elemental Warlord	Overwhelm — the task feels too large	Each small completion proves the mountain movable
12	The Veilwalker	Interdimensional Shadow	Isolation and fear of irrelevance	Each collaboration or community task tears the veil

6.4 Boss Difficulty Scaling

The assigned Super Boss has a base HP pool. This pool is modified by the AI Quality Score at each stage boundary. Higher quality work lowers the boss HP going into the final encounter. The player always wins if they complete all stages — but the fight is easier when work quality is consistently high.

BOSS ENCOUNTER DESIGN

Boss encounters are cinematic resolution sequences, not turn-based minigames. The fight is animated based on the accumulated quality score. High quality = boss withers quickly, shatters, or dissolves. Low quality = longer sequence, more dramatic corruption effects, boss retreats with residual marks on the world map.

7. Checkpoint Crossing Animation System

Six named animation patterns are defined for checkpoint crossing. Each stage uses one of these patterns, assigned to match the stage's visual setting. All patterns have a Standard (2/3) and Gold (3/3) variant.

Pattern	Standard (2/3)	Gold (3/3) Additional	Audio Cue
The Seal Break	Glowing seal cracks and shatters. Gate swings open. Party steps through.	Seal explodes in golden light. Gate transforms: iron to gilded archway. Gold marker on map.	Deep crack SFX > gate creak > flash. Gold: explosion > transformation > triumph chord.
The Rune Inscription	Stone rune tablet fills with glowing inscription. Two lines complete and tablet pulses.	All three lines complete. Tablet rises hovering. Secondary inscription appears beneath.	Ink-filling sound > soft glow pulse > stone rumble. Gold: levitation > secondary chime.
The Beacon Lighting	Watchtower beacon ignites. Visible on stage map. Light pushes back corruption nearby.	Beacon burns gold/white flame. Community feed notified as gold checkpoint event.	Flame ignition > light spread > distant flicker. Gold: colour shift > community ping.
The Bridge Repair	Broken bridge self-assembles. Path walkable. Persona physically crosses.	Bridge upgrades: stone to carved marble, rope to gilded chain. Plaque at crest.	Building SFX > click > footsteps. Gold: upgrade whoosh > plaque materialise > fanfare.
The Compass Calibration	Ground compass snaps to new heading. Fog or shadow lifts from section ahead.	Compass locks and emits directional beam to next checkpoint. Visible on community map.	Compass spin > lock click > fog dissolve. Gold: sustained beam tone > community chime.
The Ward Placement	Ward stone planted at checkpoint. Party boundary extends. Setting looks safer.	Ward grows ornate. Second stone appears. Warded area fully restored visually.	Ward plant > glow spread > boundary hum. Gold: growth creak > second stone thud.

8. Quality Scoring & AI Assessment

Every completed task submission is assessed by the AI backend. The AI does not gatekeep advancement — the user always advances if 2/3 tasks are done — but the quality of the submission is scored and contributes to the project-level Quality Score, which modifies boss difficulty.

8.1 Per-Template Quality Metrics

Template	Metric	Direction	Display Format
Venture	Valuation Score	Always increases. Quality determines magnitude.	INR proxy figure, e.g. Rs.12L pre-seed equivalent
Academic	JIF Score	Always increases. Evidence quality drives magnitude.	Decimal, e.g. 3.2
Lab	p-value	Starts high (~0.9), should decrease toward 0.05 or below.	Decimal, e.g. 0.03
Creative	Fan Score	Always increases. Release strategy and draft quality drive it.	Audience count, e.g. 2,400 fans

8.2 AI Scoring Criteria

The AI backend evaluates each submission against the checkpoint's stated Outcome. Scoring dimensions:

Dimension	Description	Score (0-3)
Completeness	Does the submission address what the task asked for?	0=missing, 1=partial, 2=adequate, 3=thorough
Specificity	Are claims concrete and grounded, or vague?	0=vague, 1=general, 2=specific, 3=highly specific
Evidence	Is the reasoning or finding supported by real data/sources?	0=none, 1=weak, 2=adequate, 3=strong
Originality	Does the submission add something beyond the task minimum?	0=none, 1=minor, 2=notable, 3=exceptional

Total score (0-12) maps to a quality tier: Low (0-4) = minimal Quality Score increment; Standard (5-8) = moderate increment; High (9-12) = full increment. The tier determines boss HP reduction per stage.

AI MODEL TIERING

Free tier: open-weight model used for scoring and AI suggestions. Explorer Pro: frontier model used for all AI interactions. The qualitative gap in feedback quality must be felt by users to drive Pro conversion. This requires A/B testing and deliberate calibration — not just a model swap.

9. Inter-Checkpoint Gameplay

Between checkpoints, after the crossing animation resolves, a short optional gameplay sequence plays — a classic arcade diversion that rewards XP and re-energises the user before the next task block.

9.1 Henchmen Encounters

Small enemies spawn on the path between checkpoints, thematically linked to the stage monster. The user can tap/click to defeat each one for a small XP bonus. They auto-resolve if the user does not interact (run away, dropping a smaller reward).

Stage	Henchman Type	XP Per Defeat	Mechanic
Venture: Ideation	Fog Wisps	+3 XP	Tap to clear
Venture: Research	Confusion Sprites	+4 XP	Quick-tap sequence
Venture: Validation	Echo Shades	+5 XP	Dodge + tap combo

9.2 Treasure Chests

20% probability per inter-checkpoint segment. Opening awards one of:

- XP Cache — Flat XP bonus (25-75 XP depending on stage)
- Flare Charge — A free Flare to broadcast a help request
- Corruption Shield — Reduces corruption by 5% immediately, slows accumulation for 48h
- Insight Fragment — Brief AI-generated insight related to the current stage topic (flavour/informational)

9.3 Journey Duration & Skip

The inter-checkpoint sequence plays for 15-30 seconds. A Skip button is always available. Skipping plays the crossing animation but omits henchmen/chest content. Already-collected XP is retained; only the journey bonus is foregone. This ensures the mechanic feels optional, not punitive.

10. Universal & Specific Tools

10.1 Universal Tools — Available on All Project Types

Tool	Description
Flare	Broadcast a help request to the community. Costs zero points to fire.
Contribution	Required at every CP and stage completion. Formats: text (50+ words), audio (30s+), video (60s+), image, or file (Excel, PPT, PDF, Doc). Auto-shared to project feed at milestones.
Kanban / To-Do	Task management board with custom columns for the project team.
Calendar	Timeline and milestone scheduling tool shared with collaborators.
Group Chat	Real-time text chat for project collaborators.
Video Call / Stream	Live video sessions within the project team.

10.2 Task-Specific Tools

Tool	Triggered By	Description
Write Tool	Write tasks	Rich text editor with markdown, AI assist, word count
Table Builder	Table tasks	Structured spreadsheet-style table with labelled columns
Map / Canvas	Map/Canvas tasks	Freeform whiteboard — post-its, diagrams, arrows, images
Survey Builder	Survey tasks	Create, distribute, collect surveys. Responses visualised.
Poll Tool	Poll tasks	Quick 2-4 option polls broadcastable to community
Link Collector	Link tasks	URL input with auto-preview and annotation
Upload Tool	Upload tasks	File upload: PDF, PPT, XLS, DOC, images, video, audio
Self-Report	Self-report tasks	Guided text form for personal confirmations and reflections
Journal	Journal tasks	Private running log; entries can be selectively shared

11. Levels & XP Progression

The platform has 50 levels across 5 phases. The first 6 levels are task-gated tutorial levels. From Level 7 onward, progression requires both a points threshold and specific qualifying actions.

11.1 Phase Overview

Phase	Levels	Cumul. Points	Timeframe (Scenario A)	Feel
Tutorial	1-6	300	Weeks 0-1	Task-gated discovery. No grind. Onboarding disguised as gameplay.
Early	7-15	5,000	Months 0-3	1 level every 1-2 weeks. Momentum building.
Mid	16-28	33,600	Months 3-22	1 level every 3-5 weeks. Real contributor identity.
Senior	29-39	84,200	Months 22-54	1 level every 6-8 weeks. Platform veteran.
Mentor	40-50	159,000	Years 4-8	1 level every 8-12 weeks. Community leader. Unlocks mentorship role.

11.2 Selected Level Milestones

Level	Title	Phase	Requirements to Advance	Pts Floor
1	Newcomer	Tutorial	Create account Upload profile photo Add 2 skills + 1 industry	Task-gated
2	Explorer	Tutorial	Browse idea feed Like 3 ideas Leave first comment	Task-gated
3	Thinker	Tutorial	Create first idea Fill title, description, type Complete first checkpoint	Task-gated
4	Connector	Tutorial	Send first collaboration invite Follow 3 users Earn 50 pts	50
5	Contributor	Tutorial	Complete 2 checkpoints Earn 150 pts Fire one Flare	150
6	Initiator	Tutorial	Complete first stage Earn 300 pts Have 1 collaborator join	300
9	Surveyor	Early	1,350 pts Left 10 total comments Respond helpfully to 1 Flare	1,200
12	Artisan	Early	3,400 pts Earned at least 1 gold checkpoint	3,000
15	Strategist	Early-	6,800 pts 1 idea at Stage 6 3 gold	5,000

Level	Title	Phase	Requirements to Advance	Pts Floor
		Mid	checkpoints	
20	Architect	Mid	19,500 pts 5 collaborators across ideas 8 gold checkpoints	12,000
28	Orchestrator	Mid-Senior	65,500 pts 1 full idea lifecycle complete 20 gold checkpoints	33,600
40	Mentor	Mentor	143,500+ pts Unlocks: accept mentees, curate ideas, earn mentee XP	~120,000
50	Visionary	Mentor	159,000+ pts 100 gold checkpoints Hall of fame status	159,000

11.3 Level-Up Animation Spec

1. Screen edge burst in accent purple — 0.3s
2. Level number counter spins from old to new — 0.5s with bounce easing
3. New title fades in with serif font, gold glow — 0.4s
4. Phase transition levels (6>7, 15>16, 28>29, 39>40) additionally show a world-map unlock animation — 1.2s
5. Any new abilities or tool unlocks displayed as floating cards — 0.8s
6. Short fanfare audio plays — 2s total sequence (skippable after 0.5s)

12. Achievement & Badge System

The platform has 62 badges across 7 categories. Badges are earned through specific actions, irrespective of level. They appear in the user's public profile and are animated on award.

12.1 Badge Category Summary

Category	Count	Rarity Mix	Role
Onboarding	8	Common, Uncommon	Early wins. All earnable in first week. Tutorial-linked.
Idea Milestones	18	Common through Legendary	Core progression rewards. First CP to 100 gold CPs.
Community	12	Common through Epic	Social layer. Giving and receiving feedback, collaboration.
Consistency	8	Common, Uncommon, Rare, Epic	Streak and return rewards.
Hidden	8	Hidden (all)	Surprise discoveries. Reward unusual or unexpected behaviour.
Aspirational	8	Legendary (all)	Visible to all, earnable by very few. Hall of fame tier.

12.2 Selected Badges by Rarity

Badge Name	Rarity	Category	Requirement
First Light	Common	Onboarding	Create your account
The Face Behind the Name	Common	Onboarding	Upload a profile photo
Marked by Trade	Common	Onboarding	Add at least 2 skills and 1 industry
The Seedling	Uncommon	Onboarding	Create your first idea
Gate Crossed	Uncommon	Onboarding	Complete all tutorial levels (1-6)
The First Checkpoint	Common	Idea Milestones	Complete your first checkpoint on any idea
Gilded	Uncommon	Idea Milestones	Complete your first gold checkpoint (all 3 tasks)
Stage Clear	Common	Idea Milestones	Complete your first full stage on any idea
The Heartland	Rare	Idea Milestones	Complete Stage 5 (hardest stage) on any idea

Badge Name	Rarity	Category	Requirement
The Launcher	Rare	Idea Milestones	Complete the Launch stage on any Venture idea
The Full Circle	Epic	Idea Milestones	Complete a full idea lifecycle — all stages
The Gilded Path	Epic	Idea Milestones	Full lifecycle with gold CPs at every stage's final checkpoint
The Polymath	Rare	Idea Milestones	Complete at least one idea in each of the four templates
Century	Legendary	Idea Milestones	Earn 100 gold checkpoints across any ideas
The Catalyst	Rare	Community	Have 5 reviews you gave marked as helpful by idea owners
The Trusted Voice	Rare	Community	Give 50 upvoted reviews or comments total

12.3 Badge Award Animation

7. Interrupt flash — brief white flash overlay (0.1s) signals something happened
8. Badge materialises — drops from top of screen, 200% scale, bounces to 100% — 0.6s
9. Reveal card — badge name, tagline, rarity tier shown in auto-dismissing panel after 4s or on tap — 0.4s fade in

Legendary badges add a full-screen gold particle burst before the interrupt flash, and the reveal card persists until manually dismissed.

13. Points Economy

Points (XP) are the universal currency of progression. Anti-farming caps apply to all social actions. The economy is designed so genuine project work accounts for the majority of points earned — not social gaming.

13.1 Key Actions & Point Ranges

Action	Min Pts	Max Pts	Anti-Farm Rule
Like / upvote an idea	1	3	Capped at 20 likes given per day
Leave a comment	2	8	No points if comment is under 10 words
Create a new idea	5	15	Reward comes from progress, not creation volume
Complete a checkpoint (2/3 tasks)	20	80	Scales with stage number. Must be genuine advance.
Complete a gold checkpoint (3/3 tasks)	40	160	2x checkpoint value for that stage
Idea advances to next stage	50	200	Scales with stage. Only on genuine completion.
Idea reaches Launch stage	150	150	Flat landmark bonus. One-time per idea.
Idea reaches Scale stage	300	300	Flat apex bonus. One-time per idea.
Send a collab invite (accepted)	10	20	Only when invite is accepted, not just sent
Respond helpfully to a Flare	15	20	Only when sender marks as helpful. 1 response per flare.
Your review helps idea advance	25	75	Stage-scaled. Requires idea owner to mark as helpful.
Complete weekly featured quest	50	150	Resets Monday. Cannot be completed retroactively.
Finish top 5 of league (weekly)	75	200	Based on weekly points only, not lifetime total.
Accept a new mentee (Lv 40+)	30	30	Must remain active with mentee for 2 weeks to receive.
Mentee completes a checkpoint	10	40	Passive — rewards sustained mentorship

FLARE ECONOMY

Firing a Flare costs zero points and earns zero points. Points only flow to the responder when the sender marks a response as helpful. This prevents abuse while incentivising community help. Later-stage flares earn

more (up to 20 pts) because the problems are harder.

14. Animation & Audio Design System

14.1 Animation Timing Standards

Animation Type	Duration	Easing / Notes
Panel open / close	280ms	cubic-bezier(0.4, 0, 0.2, 1). Framer Motion AnimatePresence.
Stage biome transitions	800ms	Crossfade with Phaser camera scroll
Checkpoint crossing sequences	1.5-3.5s	Varies by pattern type (see Section 7)
Boss entrance / exit cinematic	2-4s	Unskippable for Super Boss
Task completion checkmark	200ms	Spring bounce
XP gain counter tick-up	600ms	Ease-out
Level-up full sequence	~2s	Skippable after 0.5s
Badge award full sequence	4s	Legendary cannot be skipped
Corruption bar fill	Real-time	No animation — smooth continuous fill
Corruption threshold breach	400ms	Pulse on HUD border

14.2 Audio Architecture (Howler.js)

Category	Sound Design	Loop
Stage Ambience	Each biome has a distinct ambient loop. Village: bustle + birds. Forest: wind + insects. Mine: drips + rumble. Harbour: waves + rope. Capital: distant crowd.	Loop
Checkpoint SFX	6 patterns x 2 variants (standard / gold). Short one-shot SFX that must feel satisfying without being intrusive.	One-shot
Corruption Ambient	Drone that fades in as meter rises. Procedurally mixed: louder + lower pitch as % increases. 5 threshold layers stacked in Howler.	Layered loop
Level-Up / Badge	Distinct fanfares per event type. Level-up: ascending arpeggio with percussion. Badge: shorter, variant per rarity tier. Legendary: full chord swell 2s.	One-shot
Boss Encounter	Each of the 12 Super Bosses has a distinct 4s entrance theme. Stage monsters share a template-level theme with a per-stage variant.	One-shot
UI Actions	Soft, consistent click, hover, confirm, error SFX. Muted relative to world audio.	One-shot

AUDIO GLOBAL CONTROLS

Persistent audio settings in HUD: Master Volume | Music Volume | SFX Volume. All default ON. Muting master kills all audio. Settings persist in localStorage across sessions. Audio initialisation is deferred until first user interaction to comply with browser autoplay policies.

15. Open Technical Considerations & Pending Decisions

Item	Decision Required	Recommendation
Photo-to-Pixel Pipeline	Client-side (Canvas + WebGL) vs. server-side (Pillow + dithering script).	Server-side processing with client-side preview for consistent quality. Note latency impact.
AI Quality Score Calibration	The gap between open-weight and frontier model scoring must be felt by users to drive Pro conversion.	Dedicated A/B testing sprint. Deliberate calibration, not just model swap.
Boss Difficulty Scaling Formula	Exact formula for mapping cumulative Quality Score to boss HP reduction needs tuning.	Requires playtesting data. Too steep = boss trivial. Too flat = quality feels irrelevant.
Inter-CP Sequence Architecture	Henchmen tapping and chest discovery need to be Phaser scenes, not React components.	Clean event bridge between React panels and Phaser canvas required before build.
Community Visibility of Gold CPs	Gold CP completions should broadcast to a community feed — push notification, feed item, or in-world map event not yet specified.	Feeds team decision required. Recommend in-world map event + feed item combined.
Contribution Verification	Minimum word/length rules exist. AI plagiarism or AI-generation detection at contribution layer not yet specified.	Policy decision required. Minimum viable: word count + image hash uniqueness check.

15.1 HUD Always-Visible Elements — Final List

Element	What It Shows
XP Bar	Current points toward next level, with numeric current/target display
Level Number	Current level (1-50) with title name on hover
Stage Name	Current stage of the active project in high-fantasy lore language
Checkpoint Progress	e.g. 3/7 — checkpoints cleared / total in current stage
Streak Counter	Consecutive days of activity on the platform
Corruption Meter	Project corruption percentage with colour-coded fill
Quality Score	Current project-level quality metric (Valuation / JIF / p-value / Fan Score)
Audio Toggle	Master mute/unmute with volume slider on expand

